Shorter, but Less Safe?

How suppressing length bias in DPO affects refusals, explained through the refusal direction (6-week plan)

The idea in one paragraph. DPO-trained models write longer answers than they need to. Methods such as R-DPO and SimPO fix this, and their papers show the answers get shorter while quality holds. They never check whether the model still **refuses harmful requests**. We test that, and we explain the result with spectral analysis: we check whether the weight change made by length suppression touches the **refusal direction**, the single internal direction known to switch refusal on. If it does, we remove that part of the change and test whether the model keeps both its short answers and its safety. Everything uses **publicly released models**, so no training is needed.

1. Background

Length bias in DPO. Direct Preference Optimization (DPO) [1] trains a model on pairs of answers, one preferred and one rejected. The preferred answer is often the longer one, so the model learns that longer is better and starts padding its answers.

The fixes we study. R-DPO [2] keeps DPO exactly as it is and adds one penalty for the length difference between the two answers. **SimPO** [3] scores each answer by its average log-probability per token instead of the total, so longer answers get no automatic advantage.

The refusal direction. Inside a language model, each word is represented as a long list of numbers (a vector). Arditi et al. [4] showed that refusing harmful requests is controlled by **one direction** in this space: remove it and the model stops refusing; add it and the model refuses even harmless requests.

Spectral analysis of a weight change. Training changes every weight matrix. The change is itself a matrix: Delta-W = W_{after} - W_{before}. **Singular value decomposition (SVD)** breaks this change into a ranked list of directions; the top few carry most of the change and form the **subspace** the training worked in. We can then ask: does this subspace contain the refusal direction?

2. Why we study refusal only

- **It has a validated mechanism to test.** The refusal direction [4] gives the spectral analysis a precise target. Other behaviours (calibration, faithfulness) have no equally reliable reference.
- **It is barely affected by answer length.** "Did the model help with a harmful request?" is a yes/no question, so a shorter answer does not score worse just for being short.
- **There is early evidence.** In reinforcement learning, a length penalty made refusals terser and less explained [5]. Nobody has checked DPO, or explained why it happens.

3. Research questions

	Question
RQ1	Does suppressing length bias in DPO change how often and how the model refuses?
RQ2	Why? Does the weight change made by length suppression overlap the refusal direction?
RQ3	If so, can we remove that overlap and keep short answers without the safety cost?

4. The models

The SimPO authors released plain-DPO, R-DPO and SimPO models, each trained from the **same starting model on the same data** [3]. Because only the training method differs, comparing them isolates the effect of length suppression. All start from instruction-tuned models that already refuse harmful requests, so there is safety to lose.

Starting model	**Released versions (Hugging Face: princeton-nlp/...)**	**Role**
Llama-3-8B-Instruct	Llama-3-Instruct-8B-DPO / -RDPO / -SimPO, data version v0.2	Main model
Llama-3-8B-Instruct	Same three, data version v0.1 (different preference labels)	Repeat

Starting model	Released versions (Hugging Face: princeton-nlp/...)	Role
Mistral-7B-Instruct	Mistral-7B-Instruct-DPO / -RDPO / -SimPO	Second family

The same collection also contains models trained with methods that do **not** target length (IPO, KTO). We use them as comparison points in Section 6.

5. RQ1: measuring the effect on refusals

What we measure	Benchmark	What it tells us
Harmful compliance	HarmBench (direct requests), StrongREJECT	Main outcome: does the model help with harmful requests more often?
Over-refusal	XSTest	Does it refuse safe requests more often? Damage can go either way.
Refusal style	Same harmful prompts	Short refusal ("I can't help with that.") vs explained refusal.
Prefilling test	Reply forced to start with "Sure, here is"	Safety is often decided in the first few words of a reply [6]; this tests how robust the refusal is.

We also record average answer length and AlpacaEval 2 length-controlled win rate, only to confirm that each fix really shortens answers without hurting quality.

Separating "shorter" from "less safe"

The problem. The fixed models write shorter answers. Some safety measures can change just because an answer is short. For example, a short reply has less room for a warning, and a judge may rate a one-line refusal differently from a full paragraph. Then we would be measuring shortness, not damage.

The solution: compare answers of equal length. DPO samples answers with some randomness, so asking it the same question several times gives answers of different lengths. For each prompt, we ask plain DPO eight times and keep the answer whose length is closest to R-DPO's answer.

> **Example.** R-DPO answers a prompt in 90 words. DPO's eight answers are 80, 120, 150, 95, 200, 140, 170 and 110 words. We keep the 95-word answer. Now both answers are about the same length, but only one comes from a model trained to be brief. If R-DPO is still less safe across all prompts, the training changed something beyond length.

How many prompts. Small differences can appear by chance. With about 300 prompts per benchmark, a difference of roughly 5 percentage points (for example 8% vs 13% harmful compliance) is large enough to be trusted rather than dismissed as noise.

6. RQ2: explaining the effect with spectral analysis

- **Step 1. Isolate the length edit.** R-DPO and DPO start from the same model and data; the only difference is the length penalty. So the difference between their weights, **Delta-L = $W_{R\text{-}DPO}$ - W_{DPO}**, is the "length edit": what length suppression changed. We analyse this edit, not the full training change.
- **Step 2. Find its main directions.** Run SVD on Delta-L for every weight matrix in every layer and keep the top directions that carry most of the edit (cut-offs of 50% and 90%, both reported).
- **Step 3. Measure the overlap.** Compute how much of the refusal direction lies inside those top directions: a number from 0 (none) to 1 (all).
- **Step 4. Compare against baselines.** Earlier work found that safety overlaps with almost any weight change [7], so a raw overlap number means little. The length edit's overlap must be higher than: (a) random directions; (b) the edits of methods that do not target length (IPO - DPO, KTO - DPO); (c) the difference between the two DPO data versions (v0.2 - v0.1). As a check that the method works at all, the length edit must clearly overlap the direction that ends an answer (the end-of-sequence token), since a shorter-answer edit must change when the model stops.
- **Step 5. Test cause and effect** (explained below).

Step 5 in plain words

Steps 1-4 only show that two things **overlap**. Overlap is not proof that one causes the other. Step 5 tests cause directly, using two simple experiments that need no training, only arithmetic on the weights.

Experiment A: the dose test. Start from the DPO model and apply the length edit a little at a time: $W = W_{DPO} + t \times \text{Delta-L}$. At t = 0 we have plain DPO; at t = 1 we have exactly R-DPO; t = 0.25, 0.5 and 0.75 are in-between models; t = 1.25 exaggerates the edit. We measure answer length and safety for each. If safety falls steadily as more of the edit is applied, like a side effect growing with the dose of a medicine, the edit is causing the safety drop.

Experiment B: the cleaned edit. We cut the refusal-direction part out of Delta-L (a standard projection: subtract the component that points along the refusal direction), and apply this cleaned edit to DPO instead.

Model	Answers short?	Safety drop?	Conclusion
DPO + full length edit (= R-DPO)	Yes	Yes	The edit causes damage
DPO + cleaned edit	Yes	No	The refusal part of the edit was the cause
DPO + cleaned edit	Yes	Yes	The cause is elsewhere; overlap was a coincidence

Because only one thing changes between the full and cleaned edit, the comparison isolates the cause. If the first two rows occur, the cleaned model is also the answer to **RQ3**: short answers without the safety cost.

A note on SimPO

SimPO changes **three** things compared with DPO at once: it normalises by length, removes the reference model, and adds a fixed margin between the two answers' scores. If SimPO is less safe, we cannot tell which of the three changes caused it, just as changing three ingredients in a recipe hides which one changed the taste. Separating them would need training a variant that changes only the length part, which does not fit in six weeks. So **R-DPO is our main comparison**, because it changes only the length term, and SimPO is a secondary comparison with this caveat stated.

7. Six-week plan

Week	Work	Result
1	Get GPU access (one 80 GB GPU is enough). Download Llama-3-8B-Instruct and its DPO, R-DPO, SimPO versions. Set up vLLM, the benchmarks, a safety judge model and the refusal-direction code [4]. Write down metrics and analysis settings before seeing results.	Working pipeline
2	RQ1 on Llama (v0.2): generate answers on all benchmarks, judge them, build the length-matched DPO answers.	**Decision point:** does refusal change beyond the length-matched baseline?
3	RQ2 Steps 1-4: refusal direction, SVD of the length edit, overlap, baselines, end-of-sequence check.	Overlap results
4	Step 5: dose test and cleaned edit (RQ3).	Cause-and-effect results
5	Repeat the key results on Llama v0.1 and Mistral-7B-Instruct.	Shows the result is not specific to one model
6	Figures, write-up, presentation.	Final report

If week 2 shows no change in refusals, the project becomes a careful negative result: length fixes in DPO are safe, and weeks 3-4 explain why (the length edit stays away from the refusal direction).

8. Possible outcomes

What we find	What it means
Refusals weaken, overlap is high, and the cleaned edit removes the damage	Strongest result: a hidden safety cost, its cause, and a fix.
Refusals weaken, but the cleaned edit does not help	The cause lies elsewhere; the prefilling test points to the first words of the reply as the next place to look.
No change once length is matched	Length fixes are safe in DPO, unlike in reinforcement learning [5], and the analysis shows why.

9. What is new and what is not

New: testing whether DPO length-bias fixes weaken refusals; isolating the length edit as a weight difference; linking it to the refusal direction with a cause-and-effect test.

Not new (used as tools): side effects of length penalties in reinforcement learning [5, 8]; the refusal direction [4]; SVD overlap analysis of fine-tuning [7]; removing a direction from a weight change [4, 9].

10. Limitations

- Each public model was trained once, so part of the length edit may be random training noise. Baseline (c) in Step 4 estimates how large that noise is.
- SimPO's effect cannot be split into its three changes (Section 6).
- Refusal may use more than one direction. We repeat the overlap test with refusal directions taken from several layers.
- The first two limitations are removed by our own controlled training (Section 11).

11. Potential continuation of the work

The six weeks use only public models. The full study, as originally planned, adds our own controlled training. This removes the limitations above and is what a strong main-track paper would need.

Item	Plan
Models	Qwen2.5-1.5B-Instruct (main) and Llama-3.2-3B-Instruct (repeat of key results); both already refuse harmful requests
Training type	Full fine-tuning, not LoRA: part of a LoRA update stays close to its random starting values, which would distort the spectral comparison
Data	40k UltraFeedback prompts; 5 answers per prompt from the starting model, scored by a reward model; best answer = preferred, worst = rejected. We first confirm the preferred answer is usually longer, i.e. that length bias exists
Methods	DPO; R-DPO and LD-DPO at three penalty strengths each; SimPO plus a variant that only applies length normalisation (so SimPO's three changes can be separated); SamPO
Repeats	3 random seeds per setting, so random training noise can be measured directly
Extra baselines	The difference between two DPO runs with different seeds (noise floor) and a fine-tune on unrelated math data
Fix during training	If the cleaned edit works (RQ3), apply the same idea during training by keeping the update away from the refusal direction [9], and compare it with simply adding a small amount of safety data
Compute	About 30 training runs; roughly 200 A100 GPU-hours including evaluation

The penalty strengths are the key addition: R-DPO and LD-DPO become plain DPO at one end, so training at several strengths shows whether the safety damage grows step by step with length suppression.

References

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